

# Seonhong Min

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Research Focus: FHE, MPC

Conferences: 11 | Journals: 2 | Preprints: 2 | Reviewer: TCS, DCC

## RESEARCH INTERESTS

My main research topics are Fully Homomorphic Encryption (FHE) and Multi-Party Computation (MPC). I specialize in enhancing the performance and functionality of a variety of FHE schemes.

## EDUCATION

**Seoul National University** Mar. 2022 – Present  
Integrated M.S./Ph.D. in Computer Science & Engineering  
Advisor: Prof. Yongsoo Song

**Seoul National University** Mar. 2018 - Feb. 2022  
B.S. in Mathematics

**Daegu Science High School** Mar. 2015 – Feb. 2018

## PUBLICATIONS

Authors are listed in alphabetical order by last name, unless an asterisk(\*) is indicated.

### Conferences

**C11 2025/2057**  
**Distributed Key Generation for Efficient Threshold-CKKS**

*Seonhong Min, Guillaume Hanrot, Jai Hyun Park, Alain Passelègue, Damien Stehlé*  
ACM CCS 2026

**C10 2026/322**  
**Multi-key Fully Homomorphic Encryption with Non-Interactive Setup in the Plain Model**  
*Seonhong Min, Jeongeun Park, Yongsoo Song*  
IACR CRYPTO 2026

**C9 2025/1804**  
**\*HERDS: Multi-Key Fully Homomorphic Encryption with Sublinear Bootstrapping**  
*Binwu Xiang, Seonhong Min, Intak Hwang, Zhiwei Wang, Haoqi He, Yuanju Wei, Yi Deng, Jiang Zhang, Kang Yang, Yu Yu*  
IACR EUROCRYPT 2026

**C8 2025/382**  
**On the Security and Privacy of CKKS-based**

### Homomorphic Evaluation Protocols

*Intak Hwang, Seonhong Min, Jinyeong Seo, Yongsoo Song*  
IACR ASIACRYPT 2025

**C7 2024/2032**

**Carousel: Fully Homomorphic Encryption with Bootstrapping over Automorphism Group**  
*Intak Hwang, Seonhong Min, Yongsoo Song*  
IACR ASIACRYPT 2025

**C6 2025/216**

**Practical (Malicious) Circuit Privacy / Sanitization for TFHE**  
*Intak Hwang, Seonhong Min, Jinyeong Seo, Yongsoo Song*  
ACM CCS 2025

**C5 2025/1255**

**Efficient Full Domain Functional Bootstrapping from Recursive LUT Decomposition**  
*Intak Hwang, Shinwon Lee, Seonhong Min, Yongsoo Song*  
SAC 2025

**C4 2024/1502**

**MatriGear: Accelerated Authenticated Matrix Triple Generation with Scalable Prime Fields via Optimized HE Packing**  
*Hyunho Cha, Intak Hwang, Seonhong Min, Jinyeong Seo, Yongsoo Song*  
IEEE S&P 2025

**C3 2025/429**

**\*Enhanced CKKS Bootstrapping with Generalized Polynomial Composites Approximation**  
*Seonhong Min, Joon-woo Lee, Yongsoo Song*  
ACM ASIACCS 2025

**C2 2023/958**

**Faster TFHE Bootstrapping with Block Binary Keys**  
*Changmin Lee, Seonhong Min, Jinyeong Seo, Yongsoo Song*  
ACM ASIACCS 2023

**C1 2022/1460**

**Towards Practical MK-TFHE: Parallelizable, Key-Compatible, Quasi-Linear Complexity**  
*Hyesun Kwak, Seonhong Min, Yongsoo Song*  
IACR PKC 2024

### Journals

J2 2024/1534

**More Efficient Lattice-based OLE from Circuit-private Linear HE with Polynomial Overhead**

*Leo de Castro, Duhyeong Kim, Miran Kim, Keewoo Lee, Seonhong Min, Yongsoo Song*

ACR COMMUNICATIONS IN CRYPTOLOGY

J1 2406.14372

**\*Ring-LWE based encrypted controller with unlimited number of recursive multiplications and effect of error growth**

*Yeongjun Jang, Joowon Lee, Seonhong Min, Hyesun Kwak, Junsoo Kim, Yongsoo Song*

IEEE TRANS. ON CONTROL OF NETWORK SYSTEMS

**Preprints**

P2 2025/203

**Ciphertext-Simulatable HE from BFV with Randomized Evaluation**

*Intak Hwang, Seonhong Min, Yongsoo Song*

P1 2024/181

**Functional Bootstrapping for Packed Ciphertexts via Homomorphic LUT Evaluation**

*Dongwon Lee, Seonhong Min, Yongsoo Song*

## PRESENTATIONS

- **Enhanced CKKS Bootstrapping with Generalized Polynomial Composites Approximation** - AsiaCCS 2025
- **Faster TFHE Bootstrapping with Block Binary Keys** - FHE.org Meetup ([Video](#))
- **Faster TFHE Bootstrapping with Block Binary Keys** - AsiaCCS 2023
- **Functional Bootstrapping for Packed Ciphertexts via Homomorphic LUT Evaluation** - FHE.org Meetup ([Video](#))
- **Towards Practical MK-TFHE: Parallelizable, Key-Compatible, Quasi-Linear Complexity** - PKC 2024 ([Video](#))

## POSTERS

- **Practical MK-TFHE: Parallelizable, Key-Compatible, Quasi-Linear Complexity** - FHE.org 2023
- **Carousel: Blind Rotation Over the Automorphism Group** - FHE.org 2024
- **Practical Sanitization for TFHE** - FHE.org 2024

- **MatriGear: Accelerated Authenticated Matrix Triple Generation with Scalable Prime Fields via Optimized HE Packing** - FHE.org 2024

## HONORS & AWARDS

- **Korea Cryptography Contest 2025** - 3rd Place, 4th Place
- **Korea Cryptography Contest 2024** - 3rd Place
- **Korea Cryptography Contest 2023** - 4th Place

## EXPERIENCES

FHElab Lyon (Cryptolab Inc.) (Intern) 2024.11 – 2025.02

## OPEN-SOURCE REPOSITORIES

- **Multi-key TFHE: SNUCP/MKTFHE**
- **Carousel.jl: SNUCP/Carousel.jl**
- **HIENAA.jl: [snu-lukemin/HIENAA.jl](#)**
- **Ciphertext Simulatable BFV: SNUCP/simct**
- **CRS-less MKFHE: SNUCP/crsless**

## ACADEMIC SERVICES

**Reviewer at TCS** (Theoretical Computer Science), DCC (Designs, Codes and Cryptography)

## SKILLS

**Languages:** Korean (native), English (fluent), Japanese (conversational)

**Programming Languages:** Julia, Python, Java, Go,  $\text{\LaTeX}$